Analytics Engineering: Session 14

Q&A & Comprehensive Review

Agenda

- Open Q&A on all course topics
- Review of challenging concepts
- Common pitfalls and misconceptions
- Project review and feedback
- Best practices recap

Open Q&A

This session is **student-driven**.

Bring your questions from Sessions 1–13:

- Concepts that were unclear
- Errors you encountered in your project
- Topics you want to explore deeper
- Certification-specific concerns

Review: The dbt Workflow

The full lifecycle in one view:

```
Sources (raw data)
  → Staging (clean, 1:1 with source)
    → Intermediate (joins, logic)
      → Marts (business-ready: dim_ / mart_)
        → Exposures (dashboards, reports)
```

Key commands at each stage:

- `dbt source freshness` → Sources
- `dbt seed` → Static data
- `dbt build` → Models + Tests + Seeds + Snapshots (in DAG order)
- `dbt docs generate` → Documentation

Common Pitfalls: Modeling

Pitfall	Solution
Joining sources directly in marts	Always go through staging first
Business logic in staging models	Staging = clean only, logic in intermediate
Aggregations causing fan-out	Pre-aggregate in intermediate (like `int_order_items_summary`)
Circular references	Redesign the DAG — data flows one way

Common Pitfalls: Testing

Pitfall	Solution
Only testing marts	Test all layers — staging PKs, intermediate joins, mart business rules
Forgetting `relationships` tests	Validate foreign keys between layers
Not using `dbt build`	`dbt build` runs tests immediately after models — catches issues early
Ignoring test failures	Fix failures before merging to main

Common Pitfalls: Configuration

Pitfall	Solution
Wrong materialization	Views for staging, tables for marts, incremental for large facts
Missing `unique_key` on incremental	Always set `unique_key` to prevent duplicates
Contracts without all columns typed	Every column needs `data_type` when contract is enforced
Snapshot without `updated_at`	Timestamp strategy requires the column to exist and be populated

Review: Certification Domains

The 8 domains tested on the exam:

#	Domain	Key Topics
1	Developing dbt models	ref, source, materializations, DAGs, Python models, grants
2	Model governance	Contracts, versions, access, deprecation
3	Debugging errors	Compiled SQL, dbt.log, YAML vs SQL errors
4	Managing pipelines	DAG failures, dbt clone, integrated tools
5	Implementing tests	Generic, singular, custom generic
6	Documentation	YAML descriptions, doc blocks, docs site

Tricky Exam Topics

Topics that students commonly find challenging:

- **Incremental strategies**: When to use append vs merge vs delete+insert
- **Snapshot config**: timestamp vs check strategy, what columns dbt adds
- **State selectors**: `state:modified` vs `state:new` vs `result:error`
- **Contract enforcement**: What happens when a contract fails at build time
- **Custom generic tests**: How they differ from singular tests
- **Exposures**: How they appear in the DAG and what `depends_on` does
- `dbt build` **vs** `dbt run`: Build includes tests, seeds, and snapshots

Project Review

Let's review your projects together:

- [ ] All staging models with sources defined
- [ ] Intermediate models isolating complexity
- [ ] Mart models (dimensions and facts)
- [ ] Tests at every layer (generic, singular, custom generic)
- [ ] Documentation on all public models
- [ ] At least one exposure defined
- [ ] At least one incremental model
- [ ] At least one snapshot
- [ ] Contracts on mart models
- [ ] `dbt build` passes with zero failures

Best Practices Recap

- [ ] **Naming**: Consistent `stg_`, `int_`, `dim_`, `mart_` prefixes
- [ ] **Structure**: Sources → Staging → Intermediate → Marts
- [ ] **Testing**: Test all primary keys and foreign keys
- [ ] **Documentation**: Descriptions for all public models and columns
- [ ] **Git**: Atomic commits, descriptive PRs, branch per feature
- [ ] **Materializations**: Views for staging, tables for marts
- [ ] **DRY**: One definition of each metric, referenced via `ref()`

What have we learned in this session

- Addressed open questions across all course topics
- Reviewed common pitfalls in modeling, testing, and configuration
- Reviewed all 8 certification domains and tricky exam topics
- Received feedback on projects
- Recapped best practices for production-ready dbt projects

Next Session: Certification Prep & Mock Exam.